

■ Neural Networks Across ML Paradigms

Neural Networks (NNs) are flexible function approximators. Depending on how they are trained — with labeled data, unlabeled data, limited labels, or through interactions — they can be applied in all Machine Learning paradigms: Supervised, Unsupervised, Semi-Supervised, and Reinforcement Learning.

1. Supervised Learning (Labeled Data)

Definition: Model learns mapping $X \rightarrow Y$ from labeled data.

NN Role: Learns hidden representations to map input to target.

Common Architectures: ANN, CNN, RNN, Transformers.

Examples: Spam classification, house price prediction, medical diagnosis.

2. Unsupervised Learning (Unlabeled Data)

Definition: Model finds hidden structure without labels.

NN Role: Learns compressed latent representations.

Common Architectures: Autoencoders, Variational Autoencoders, GANs, SOMs.

Examples: Customer segmentation, image compression, generative art.

3. Semi-Supervised Learning (Few Labels + Lots of Unlabeled Data)

Definition: Combines small labeled data with large unlabeled data.

NN Role: Pretrains on unlabeled data, fine-tunes with labels.

Common Architectures: Self-training NNs, Semi-Supervised GANs, SSL Transformers.

Examples: Medical imaging, NLP with few labels, fraud detection.

4. Reinforcement Learning (Learning by Interaction)

Definition: Agent learns via trial-and-error, guided by rewards.

NN Role: Approximates policy (actions) or value function (expected reward).

Common Architectures: Deep Q-Networks, Policy Gradient, Actor-Critic.
Examples: AlphaGo, self-driving cars, robotics.

Quick Comparison Table

Paradigm	Data Type	NN Role	Examples
Supervised	Labeled	Predict outputs	Classification, Regression, NLP
Unsupervised	Unlabeled	Learn patterns, compress, generate	Autoencoders, GANs, Clustering
Semi-Supervised	Few labels + unlabeled	Pretrain + fine-tune	SSL Transformers, Semi-GANs
Reinforcement	Reward feedback	Learn policy/value	Games, Robotics, Self-driving cars